

General equation of the second degree

* Let the second degree ~~given~~ general equation is given by,

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \longrightarrow (1)$$

Here, $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

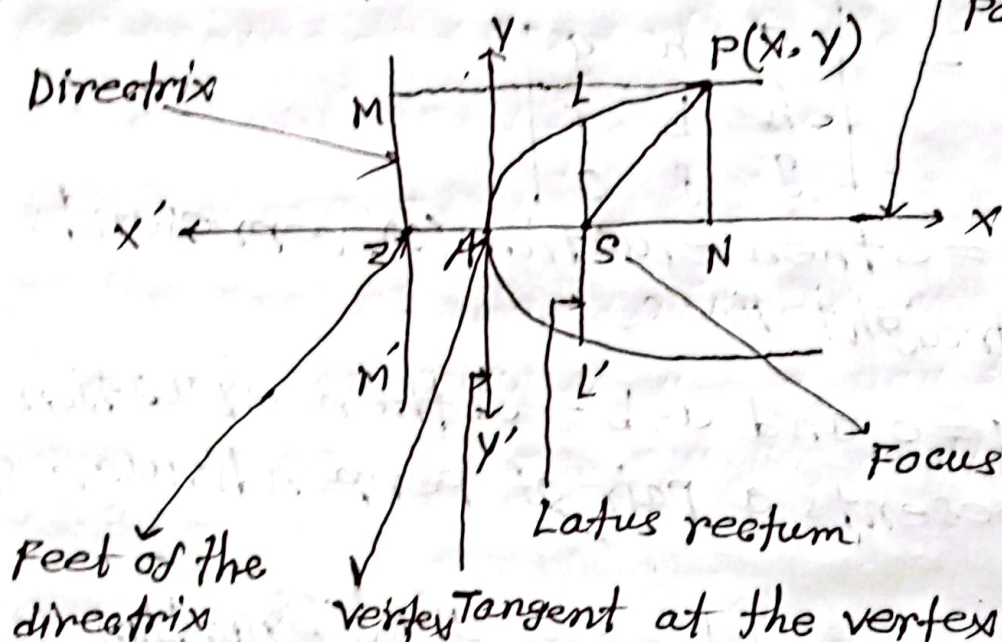
or, $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$

- i. If $\Delta = 0$, then equation (1) represents a pair of straight lines.
- ii. If $\Delta = 0$ and $a+b=0$, then equation (1) represents a pair of perpendicular straight lines.
- iii. If $\Delta = 0$ and $h^2 = ab$, then equation (1) represents a pair of parallel straight lines.
- iv. If $\Delta \neq 0$ and $ab - h^2 = 0$, then equation (1) represents a parabola.
- v. If $\Delta \neq 0$ and $ab - h^2 > 0$, then equation (1) represents an Ellipse.
- vi. If $\Delta \neq 0$ and $ab - h^2 < 0$, then equation (1) represents a hyperbola.
- vii. If $\Delta \neq 0$, $ab - h^2 < 0$ and $a+b=0$, then equation (1) represents a rectangular hyperbola.

viii. If $a \neq 0$, $h = 0$ and $a = b$, then equation (2) represents a circle.

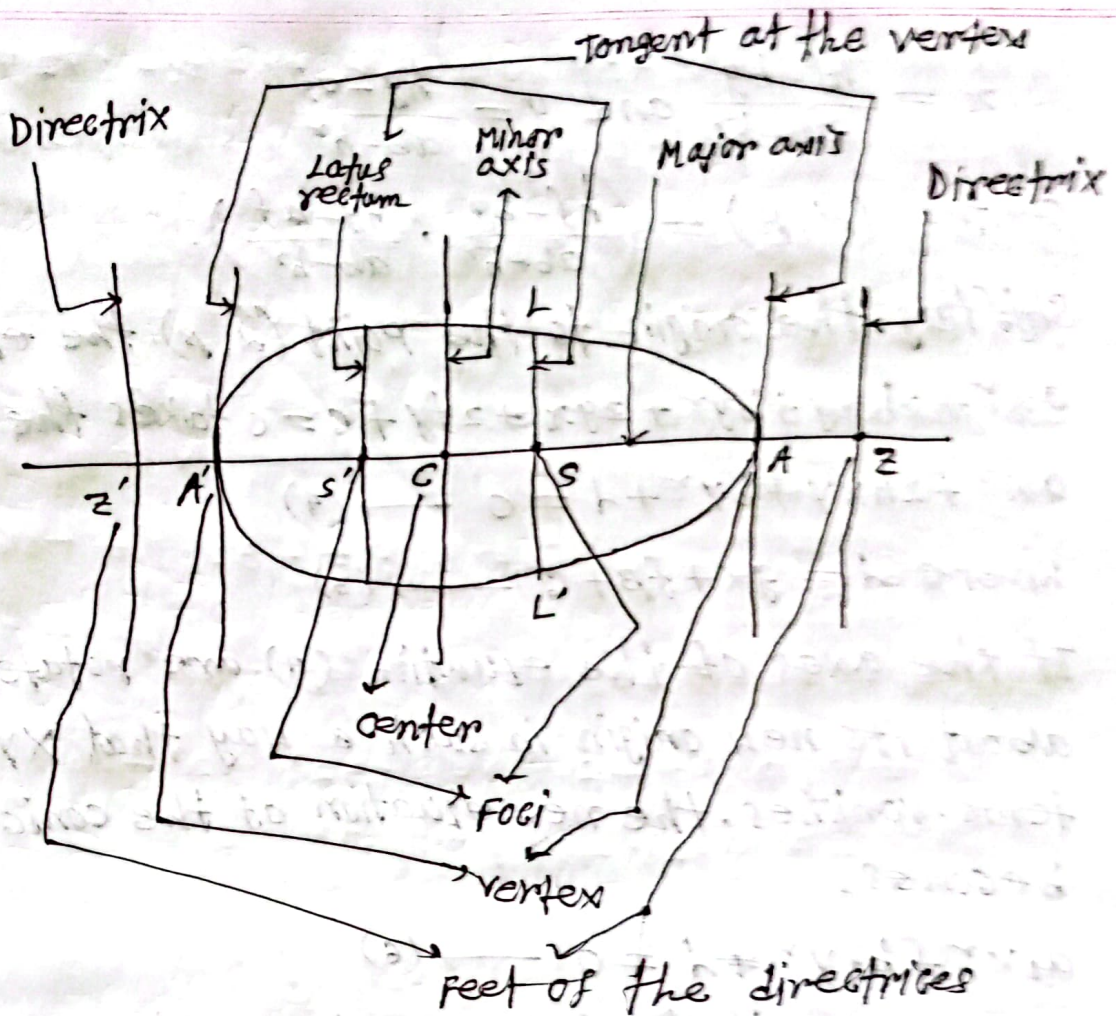
Parabola

*. Standard equation: $y^2 = 4ax$, Axis of the Parabola



- i. Vertex, $A(0,0)$.
- ii. Focus, $S(a,0)$
- iii. Equation of the axis of the parabola, $y=0$.
- iv. Equation of the latus rectum, $x=a$
- v. Length of the latus rectum, $4a$
- vi. Equation of the tangent at the vertex, $x=0$
- vii. Equation of the directrix, $x=-a$
- viii. Foot of the directrix, $Z(-a,0)$.
- ix. Co-ordinates of the end points of the latus rectum, $L(a,2a)$ and $L'(a,-2a)$.

Ellipse



*. Standard equation: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Rule: The given equation is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + G = 0 \rightarrow (1)$$

$$\text{Let, } F(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + G = 0$$

$$\therefore \frac{\partial F}{\partial x} = 2ax + 2hy + 2g = 0 \Rightarrow ax + hy + g = 0 \rightarrow (2)$$

$$\text{and } \frac{\partial F}{\partial y} = 2hx + 2by + 2f = 0 \Rightarrow hx + by + f = 0 \rightarrow (3)$$

$$\text{Solving (2) \& (3) } \Rightarrow \frac{x}{hf - bg} = \frac{y}{hg - af} = \frac{1}{ab - h^2}$$

$$\therefore x = \frac{hf - bg}{ab - h^2} \text{ and } y = \frac{hg - af}{ab - h^2}$$

$$\text{Centre, } (\alpha, \beta) = \left(\frac{hf - bg}{ab - h^2}, \frac{hg - af}{ab - h^2} \right).$$

Shifting the origin to the point (α, β) , the eqn. $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ takes the form $ax^2 + 2hxy + by^2 + d = 0 \longrightarrow (4)$

$$\text{Where } d = g\alpha + f\beta + c \longrightarrow (5)$$

If the axes of the equation (4) are rotated about its new origin in such a way that XY -term vanishes, the new equation of the conic becomes,

$$a_1x^2 + b_1y^2 + d = 0 \longrightarrow (6)$$

By the method of invariants, we have

$$a_1 + b_1 = a + b$$

$$a_1 b_1 = ab - h^2$$

Solving these equations and obtain the values of a_1 and b_1 .

Putting a_1 and b_1 , from (6) $\Rightarrow$

$$\frac{x^2}{-\frac{d}{a_1}} + \frac{y^2}{-\frac{d}{b_1}} = 1 \quad \text{where } -\frac{d}{a_1} > 0, -\frac{d}{b_1} > 0$$

Which is the standard form of an ellipse.

From (4) $\Rightarrow -\frac{a}{1} x^2 + 2\frac{h}{1} xy + \frac{b}{1} y^2 = 1$

Properties:

$\Rightarrow Ax^2 + 2Hxy + By^2 = 1 \quad \text{--- (1)}$

i. The lengths of the ^{semi-}axes are

~~$A = \pm \frac{1}{r_1}$~~ $\frac{1}{r_1} - (A+B) \frac{1}{r_2} + AB - H^2 = 0$

Find (r_1, r_2) , $r_1 > r_2$

ii. The equations of the major and minor axes are

$(A - \frac{1}{r_1})x + Hy = 0$ and $(A - \frac{1}{r_2})x + Hy = 0$

iii. Equation of the major axis

iv. " " " minor

v. Eccentricity, $e = \sqrt{1 - \frac{r_2}{r_1}}$

vi. Slope of major axis, find $(\cos \theta, \sin \theta)$

vii. ~~$d = \pm r_1 e$~~ $d = \pm r_1 e$

viii. Focus $(\alpha + d \cos \theta, \beta + d \sin \theta)$

ix. Center x. vertex xi. Foot of the director

xii. Length of the latus rectum $= \frac{2r_2}{r_1}$

Hyperbola

as same process.

Problem-1: $16x^2 - 24xy + 9y^2 - 104x - 172y + 44$ to the standard form. Find all its properties.

Solution: The given equation is,

$$16x^2 - 24xy + 9y^2 - 104x - 172y + 44 = 0 \longrightarrow (1)$$

Comparing the general equation of second degree, we get

$$a = 16, h = -12, b = 9, g = -52, f = -86, c = 44$$

$$\text{Now, } \Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= 16 \cdot 9 \cdot 44 + 2 \cdot (-86) \cdot (-52) \cdot (-12) - 16 \cdot (-86)^2$$

$$- 9 \cdot (-52)^2 - 44 \cdot (-12)^2$$

$$\neq 0.$$

$$\text{and } ab - h^2 = 16 \cdot 9 - (-12)^2 = 0.$$

Since $\Delta \neq 0$ and $ab - h^2 = 0$, so (1) will represent a parabola.

$$\text{From (1)} \Rightarrow (4x - 3y)^2 = 104x + 172y - 44 \longrightarrow (2)$$

Here, $4x - 3y$ and $104x + 172y - 44$ are not at right angles. hence let us introduce a constant λ in (2).

$$\text{From (2)} \Rightarrow (4x - 3y + \lambda)^2 = 104x + 172y - 44$$

$$+ 2\lambda(4x - 3y) + \lambda^2$$

$$\Rightarrow (4x - 3y + \lambda)^2 = (8\lambda + 104)x + (-6\lambda + 172)y + (\lambda^2 - 44)$$

The lines $4x - 3y + d = 0$ and $(8d + 104)x + (-6d + 172)y + (d^2 - 44)$ are at right angles.

$$\therefore a_1 a_2 + b_1 b_2 = 0$$

$$\Rightarrow 4(8d + 104) - 3(-6d + 172) = 0$$

$$\Rightarrow d = 2 \rightarrow (4)$$

Using (4), from (3) $\Rightarrow$

$$(4x - 3y + 2)^2 = 40(3x + 4y - 1)$$

$$\Rightarrow \left(\frac{4x - 3y + 2}{\sqrt{4^2 + (-3)^2}} \right)^2 \times 25 = 40 \left(\frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}} \right) \times 5$$

$$\Rightarrow \left(\frac{4x - 3y + 2}{5} \right)^2 = 8 \left(\frac{3x + 4y - 1}{5} \right)$$

$$\Rightarrow Y^2 = 4 \cdot 2X = 4aX, \quad a = 2, \quad Y = \frac{4x - 3y + 2}{5}, \quad X = \frac{3x + 4y - 1}{5}$$

Which is the standard form (parabola) of equation (1).

i. vertex, $(0, 0)$

$$\therefore X = 0, Y = 0$$

$$\Rightarrow 4x - 3y + 2 = 0, \quad 3x + 4y - 1 = 0$$

$$\therefore (x, y) = \left(-\frac{1}{5}, \frac{2}{5} \right)$$

ii. Focus, $(a, 0)$

$$\therefore X = a, Y = 0$$

$$\Rightarrow \frac{4x - 3y + 2}{5} = 2, \quad Y = \frac{3x + 4y - 1}{5} = 0$$

$$\Rightarrow \frac{3x + 4y - 1}{5} = 2, \quad \frac{4x - 3y + 2}{5} = 0$$

$$\Rightarrow 3x + 4y - 11 = 0, 4x - 3y + 2 = 0$$

The focus is $(x, y) = (1, 2)$

iii. Equation of the axis of the parabola, $y = 0$

$$\Rightarrow \frac{4x - 3y + 2}{5} = 0 \Rightarrow 4x - 3y + 2 = 0$$

iv. Equation of the latus rectum, $x = a$

$$\Rightarrow \frac{3x + 4y - 1}{5} = 2 \Rightarrow 3x + 4y - 11 = 0$$

v. Length of the latus rectum, $|4a| = |4 \cdot 2| = 8$

vi. Equation of the tangent at the vertex, $x = 0$

$$\Rightarrow \frac{3x + 4y - 1}{5} = 0 \Rightarrow 3x + 4y - 1 = 0$$

vii. Equation of the directrix, $x = -a$

$$\Rightarrow \frac{3x + 4y - 1}{5} = -2 \Rightarrow 3x + 4y + 9 = 0$$

viii. Foot of the directrix, $(-a, 0)$

$$x = -a, y = 0$$

$$\Rightarrow \frac{3x + 4y - 1}{5} = -2, \frac{4x - 3y + 2}{5} = 0$$

$$\Rightarrow 3x + 4y + 9 = 0, 4x - 3y + 2 = 0$$

$\therefore$ The foot of the directrix is $(-7/5, -8/5)$.

Problem-2: Find the nature of the conic $x^2 + 2xy + y^2 - 6x - 2y + 4 = 0$. Reduce it to standard form. Find all its properties.

Solution: The given equation is

$$x^2 + 2xy + y^2 - 6x - 2y + 4 = 0 \longrightarrow (1)$$

Comparing the general equation of second degree

We get $a=1, h=1, b=1, g=-3, f=-1, c=4$

$$\text{Now, } \Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= 1 \cdot 1 \cdot 4 + 2 \cdot (-1) \cdot (-3) \cdot 1 - 1 \cdot (-1)^2 - 1 \cdot (-3)^2 - 4 \cdot 1^2$$

$$= -4 \neq 0.$$

$$\text{and } ab - h^2 = 1 \cdot 1 - 1^2 = 0$$

Since $\Delta \neq 0, ab - h^2 = 0$, the given conic represents a parabola.

$$\text{From (1)} \Rightarrow (x+y)^2 = 6x + 2y - 4$$

$$\Rightarrow (x+y+\lambda)^2 = 6x + 2y - 4 + 2\lambda(x+y) + \lambda^2 \longrightarrow (2)$$

$$= (2\lambda + 6)x + (2\lambda + 2)y + (\lambda^2 - 4)$$

Let the lines $x+y+\lambda=0$ and $(2\lambda+6)x + (2\lambda+2)y + (\lambda^2-4)$ are perpendicular.

$$\therefore a_1 a_2 + b_1 b_2 = 0$$

$$\Rightarrow 1 \cdot (2\lambda + 6) + 1 \cdot (2\lambda + 2) = 0$$

$$\Rightarrow 2\lambda + 2\lambda + 8 = 0$$

$$\Rightarrow \lambda = -2.$$

Putting $\lambda = -2$, in equation (2), we get

From (2) $\Rightarrow (x+y-2)^2 = 2x-2y$

$$\Rightarrow \left(\frac{x+y-2}{\sqrt{1^2+1^2}} \right)^2 \times 2 = \frac{2(x-y)}{\sqrt{1^2+1^2}} \times \sqrt{2}$$

$$\Rightarrow \left(\frac{x+y-2}{\sqrt{2}} \right)^2 = \sqrt{2} \cdot \frac{x-y}{\sqrt{2}}$$

$$\Rightarrow y^2 = 4 \cdot \frac{\sqrt{2}}{4} \cdot x$$

$$\Rightarrow y^2 = 4 \cdot \frac{1}{2\sqrt{2}} x = 4ax$$

Here, $y = \frac{x+y-2}{\sqrt{2}}$, $x = \frac{x-y}{\sqrt{2}}$, $a = \frac{1}{2\sqrt{2}}$

Which is required standard form (parabola).

i. vertex, (0,0)

$$\therefore x=0, y=0$$

$$\Rightarrow \frac{x-y}{\sqrt{2}} = 0, \frac{x+y-2}{\sqrt{2}} = 0$$

$$\Rightarrow x-y=0, x+y-2=0$$

$$\therefore (x,y) = (1,1)$$

ii. focus, (a,0)

$$\therefore x=a, y=0$$

$$\Rightarrow \frac{x-y}{\sqrt{2}} = \frac{1}{2\sqrt{2}}, \frac{x+y-2}{\sqrt{2}} = 0$$

$$\Rightarrow x-y = \frac{1}{2}, x+y-2=0$$

$$\therefore (x,y) = \left(\frac{5}{4}, \frac{3}{4} \right)$$

iii. Equation of the axis of the parabola $y=0$

$$\text{From (2)} \Rightarrow (x+y-2)^r = 2x-2y$$

$$\Rightarrow \left(\frac{x+y-2}{\sqrt{1^2+1^2}} \right)^r \times 2 = \frac{2(x-y)}{\sqrt{1^2+1^2}} \times \sqrt{2}$$

$$\Rightarrow \left(\frac{x+y-2}{\sqrt{2}} \right)^r = \sqrt{2} \cdot \frac{x-y}{\sqrt{2}}$$

$$\Rightarrow y^r = 4 \cdot \frac{\sqrt{2}}{4} \cdot x$$

$$\Rightarrow y^r = 4 \cdot \frac{1}{2\sqrt{2}} x = 4ax$$

$$\text{Here, } y = \frac{x+y-2}{\sqrt{2}}, x = \frac{x-y}{\sqrt{2}}, a = \frac{1}{2\sqrt{2}}$$

Which is required standard form (parabola).

i. vertex, (0,0)

$$\therefore x=0, y=0$$

$$\Rightarrow \frac{x-y}{\sqrt{2}} = 0, \frac{x+y-2}{\sqrt{2}} = 0$$

$$\Rightarrow x-y=0, x+y-2=0$$

$$\therefore (x,y) = (1,1)$$

ii. focus, (a,0)

$$\therefore x=a, y=0$$

$$\Rightarrow \frac{x-y}{\sqrt{2}} = \frac{1}{2\sqrt{2}}, \frac{x+y-2}{\sqrt{2}} = 0$$

$$\Rightarrow x-y = \frac{1}{2}, x+y-2=0$$

$$\therefore (x,y) = \left(\frac{5}{4}, \frac{3}{4} \right)$$

iii. Equation of the axis of the parabola $y=0$

$$\Rightarrow x+y-2=0$$

iv. Equation of the latus rectum, $x = a$

$$\Rightarrow \frac{x-y}{\sqrt{2}} = \frac{1}{2\sqrt{2}}$$

$$\Rightarrow x - y - \frac{1}{2} = 0$$

v. Length of the latus rectum, $|4a| = |4 \cdot \frac{1}{2\sqrt{2}}| = \sqrt{2}$

vi. Equation of the tangent at the vertex, $x = 0$

$$\Rightarrow \frac{x-y}{\sqrt{2}} = 0 \Rightarrow x - y = 0$$

vii. Equation of the directrix, $x = -a$

$$\Rightarrow \frac{x-y}{\sqrt{2}} = -\frac{1}{2\sqrt{2}}$$

$$\Rightarrow x - y + \frac{1}{2} = 0$$

viii. Foot of the directrix, $(-a, 0)$

$$\therefore x = -a, \quad y = 0$$

$$\Rightarrow \frac{x-y}{\sqrt{2}} = -\frac{1}{2\sqrt{2}}, \quad y = 0 \Rightarrow \frac{x+y-2}{\sqrt{2}} = 0$$

$$\Rightarrow x - y + \frac{1}{2} = 0, \quad x + y - 2 = 0$$

$$\therefore (x, y) = \left(\frac{3}{2}, \frac{1}{2}\right)$$

Problem-3: Reduce the conic $8x^2 - 4xy + 5y^2 - 16x - 14y + 17 = 0$ to the standard form and find its some properties.

Solution: The given equation is

$$8x^2 - 4xy + 5y^2 - 16x - 14y + 17 = 0 \quad \text{--- (1)}$$

Comparing the general equation of second degree, we get $a = 8$, $h = -2$, $b = 5$, $g = -8$, $f = -7$, $c = 17$

$$\text{Now, } \Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= 8 \cdot 5 \cdot 17 + 2 \cdot (-7) \cdot (-8) \cdot (-2) - 8 \cdot (-7)^2 - 5 \cdot (-8)^2 - 17 \cdot (-2)^2$$

$$\neq 0$$

$$\text{and } ab - h^2 = 8 \cdot 5 - (-2)^2 = 36 > 0$$

since $\Delta \neq 0$, $ab - h^2 > 0$, the given conic represents an ellipse.

$$\text{Let, } F(x, y) = 8x^2 - 4xy + 5y^2 - 16x - 14y + 17 = 0$$

$$\therefore \frac{\partial F}{\partial x} = 16x - 4y - 16 = 0 \Rightarrow 4x - y - 4 = 0$$

$$\text{and } \frac{\partial F}{\partial y} = -4x + 10y - 14 = 0 \Rightarrow 2x - 5y + 7 = 0$$

$$\therefore \frac{x}{-7-20} = \frac{y}{-8-28} = \frac{1}{-20+2}$$

$$\Rightarrow (x, y) = \left(-\frac{3}{2}, 2\right)$$

$$\therefore \text{center } \left(-\frac{3}{2}, 2\right) = (\alpha, \beta)$$

Transfer the to a point $(\frac{3}{2}, 2)$, when axes being unchanged (1) can be written as,

$$8x^2 - 4xy + 5y^2 + 1 = 0 \longrightarrow (2)$$

Where, $1 = g\alpha + f\beta + c$

$$= (-8) \cdot \frac{3}{2} + (-7) \cdot 2 + 17$$

$$= -9$$

Putting $1 = -9$, from (2) $\Rightarrow$

$$8x^2 - 4xy + 5y^2 - 9 = 0$$

$$\Rightarrow 8x^2 - 4xy + 5y^2 = 9 \longrightarrow (3)$$

$$\Rightarrow \frac{8}{9}x^2 - \frac{4}{9}xy + \frac{5}{9}y^2 = 1 \longrightarrow (4)$$

$$\therefore A = \frac{8}{9}, B = \frac{5}{9}, H = -\frac{2}{9}$$

NOW, we remove xy term, then (3) $\Rightarrow$

$$a_1x^2 + b_1y^2 = 9 \longrightarrow (5)$$

We know, invariant laws

$$a_1 + b_1 = a + b = 8 + 5 = 13$$

$$a_1 \cdot b_1 = ab - h^2 = 8 \cdot 5 - (-2)^2 = 36$$

$$\text{NOW, } a_1 - b_1 = \sqrt{(a_1 + b_1)^2 - 4a_1b_1}$$

$$= \sqrt{169 - 144}$$

$$\therefore a_1 = 9, b_1 = 4 \longrightarrow (6)$$

Using (6), from (5) $\Rightarrow$

$$x^2 + \frac{y^2}{\left(\frac{3}{2}\right)^2} = 1$$

Which is required standard form (ellipse).

i. When length of the semi-axis are,

$$\frac{1}{r^4} - (A+B)\frac{1}{r^2} + (AB-H^2) = 0$$

$$\Rightarrow \frac{1}{r^4} - \left(\frac{8}{9} + \frac{5}{9}\right)\frac{1}{r^2} + \left\{\frac{8}{9} \cdot \frac{3}{9} - \left(-\frac{2}{9}\right)^2\right\} = 0$$

$$\Rightarrow 4r^4 - 13r^2 + 9 = 0$$

$$\Rightarrow (r^2 - 1)(4r^2 - 9) = 0$$

$$\therefore r_1 = \frac{3}{2}, r_2 = 1$$

$$\text{Length of major axis} = 2r_1 = 2 \cdot \frac{3}{2} = 3$$

$$\text{Length of minor axis} = 2r_2 = 2 \cdot 1 = 2$$

ii. Equation of major axis, $\left(A - \frac{1}{r_1^2}\right)(x - \alpha) + H(y - \beta) = 0$

$$\Rightarrow \left(\frac{8}{9} - \frac{4}{9}\right)\left(x - \frac{3}{2}\right) + \left(-\frac{2}{9}\right)(y - 2) = 0$$

$$\Rightarrow 4\left(x - \frac{3}{2}\right) - 2(y - 2) = 0$$

$$\Rightarrow 4x - 2y - 2 = 0$$

$$\Rightarrow 2x - y - 1 = 0$$

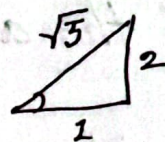
iii. Equation of minor axis, $\left(A - \frac{1}{r_2^2}\right)(x - \alpha) + H(y - \beta) = 0$

$$\Rightarrow \left(\frac{8}{9} - 1\right) \left(x - \frac{3}{2}\right) + \left(-\frac{2}{9}\right) (y - 2) = 0$$

$$\Rightarrow \left(x - \frac{3}{2}\right) + 2(y - 2) = 0$$

$$\Rightarrow x + 2y - \frac{11}{2} = 0$$

iv. Slope of major axis, $\tan \theta = 2$



$$\cos \theta = \frac{1}{\sqrt{5}}$$

$$\sin \theta = \frac{2}{\sqrt{5}}$$

v. Slope of minor axis, $\tan \theta = -\frac{1}{2}$

vi. Eccentricity, $e = \sqrt{1 - \frac{r_2^2}{r_1^2}} = \sqrt{1 - \left(\frac{2}{3}\right)^2} = \frac{\sqrt{5}}{3}$

vii. Vertices $(\alpha \pm r_1 \cos \theta, \beta \pm r_1 \sin \theta)$

$$= \left(\frac{3}{2} \pm \frac{3}{2} \cdot \frac{1}{\sqrt{5}}, 2 \pm \frac{3}{2} \cdot \frac{2}{\sqrt{5}}\right)$$

$$= \left(\frac{3}{2} \pm \frac{3}{2\sqrt{5}}, 2 \pm \frac{3}{\sqrt{5}}\right)$$

viii. Focus $(\alpha \pm r_1 e \cos \theta, \beta \pm r_1 e \sin \theta)$

$$= \left(\frac{3}{2} \pm \frac{3}{2} \cdot \frac{\sqrt{5}}{3} \cdot \frac{1}{\sqrt{5}}, 2 \pm \frac{3}{2} \cdot \frac{\sqrt{5}}{3} \cdot \frac{2}{\sqrt{5}}\right)$$

$$= \left(\frac{3}{2} \pm \frac{1}{2}, 2 \pm 1\right)$$

$$= (2, 3), (1, 1)$$

ix. Length latus rectum $\left| \frac{2r_2^2}{r_1} \right| = \left| \frac{2 \cdot 1^2}{\frac{3}{2}} \right| = \frac{4}{3}$

x. Foot of directrices $(\alpha \pm \frac{r_1}{e} \cos \theta, \beta \pm \frac{r_1}{e} \sin \theta)$

$$= \left(\frac{12}{5}, \frac{19}{5}\right), \left(\frac{3}{5}, \frac{1}{5}\right)$$

problem-4: Find the nature of the conic represented by $5x^2 + 2xy + 3y^2 + 10x - 22y + 29 = 0$ convert it into the standard ~~and~~ form and find the equation of the axes and its length if possible.

problem-5: Reduce the equation $x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0$ to the standard form and identify it.

problem-6: Find the nature of the conic $x^2 + 2xy + y^2 - 6x - 2y + 4 = 0$. Reduce it to standard form. Find vertex, focus, equation and length of latus rectum of the conic.

problem-7: Reduce the equation $8x^2 + 4xy + 5y^2 - 24x - 24y = 0$ to the standard form and find its all properties.

Circle + system of Circle

Circle: If a point moves in a plane so that its distance from a fixed point in the plane is always equal (constant), then the locus of the point is called circle. The fixed point is called centre of the circle and the constant distance is the radius of the circle.

or, The set of all points which are equidistance from a fixed point in a plane is called circle.

1. Equation of circle:

$$(x-h)^2 + (y-k)^2 = a^2$$

2. Equation of circle whose

Centre is origin: $x^2 + y^2 = a^2$

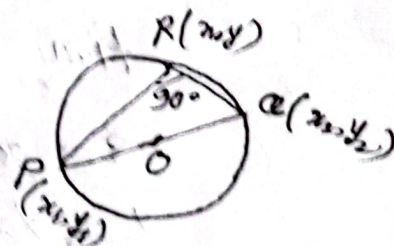
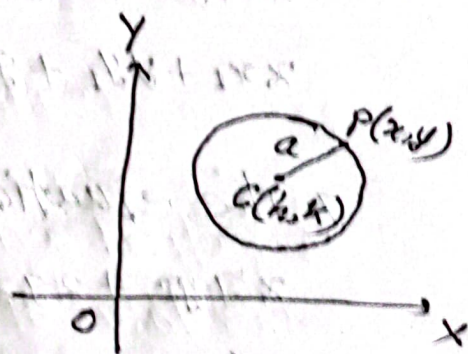
3. General equation of circle:

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Centre $(-g, -f)$, radius $= \sqrt{g^2 + f^2 - c}$

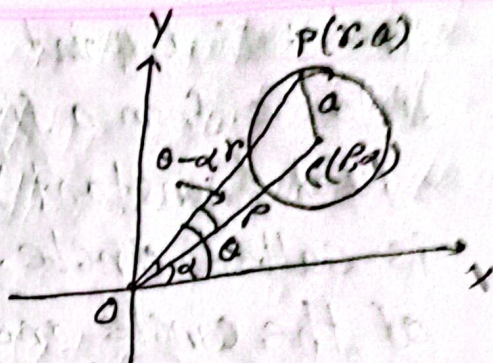
4. The equation of the circle whose diameter is the line joining the points (x_1, y_1) and (x_2, y_2) :

$$(x-x_1)(x-x_2) + (y-y_1)(y-y_2) = 0$$



5. Polar equation of circle:

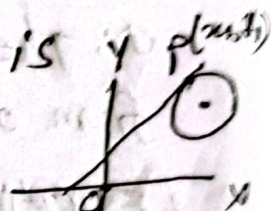
$$ar = pr + r^2 - 2pr \cos(\theta - \alpha)$$



6. The equation of the tangent to the circle:

$$x^2 + y^2 + 2gx + 2fy + c = 0 \text{ at } (x_1, y_1) \text{ is}$$

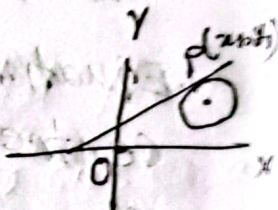
$$xx_1 + yy_1 + g(x+x_1) + f(y+y_1) + c = 0$$



7. The equation of the normal to the circle:

$$x^2 + y^2 + 2gx + 2fy + c = 0 \text{ at } (x_1, y_1) \text{ is}$$

$$y(x_1 + g) - x(y_1 + f) + fx_1 - gy_1 = 0$$



8. Length of the tangent T from an external point (x_1, y_1) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

$$T^2 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$$



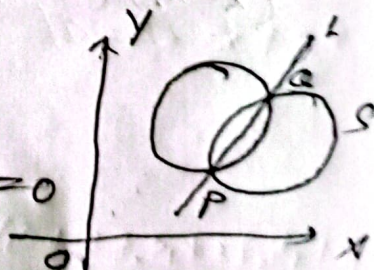
9. Equation of circles passing through the intersection points of a circle and a straight line:

$$S = x^2 + y^2 + 2gx + 2fy + c = 0$$

$$L = lx + my + n = 0$$

$$\therefore x^2 + y^2 + 2gx + 2fy + c + \lambda(lx + my + n) = 0$$

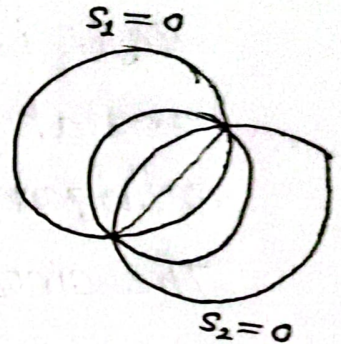
$$\Rightarrow S + \lambda L = 0 \quad ; \quad \lambda = \text{constant}$$



10. Equation of circles passing through the intersection points of two circles:

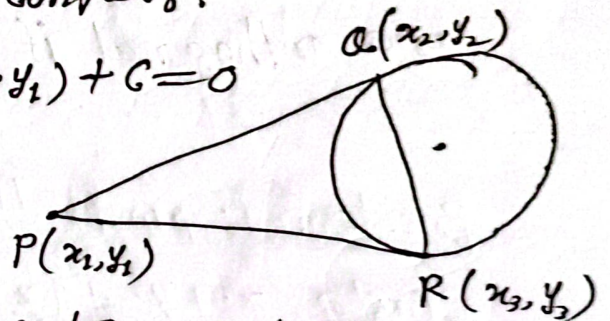
$$\therefore S_1 + \lambda S_2 = 0$$

$$\Rightarrow x^2 + y^2 + 2g_1x + 2f_1y + C_1 + \lambda(x^2 + y^2 + 2g_2x + 2f_2y + C_2) = 0$$



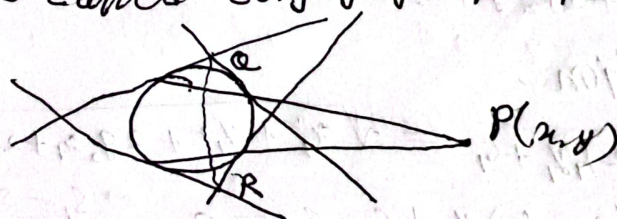
11. Equation of Chord of contact:

$$xx_1 + yy_1 + g(x+x_1) + f(y+y_1) + C = 0$$



12. The pole, polar, conjugate points of circle:

The locus of the intersection points of all pair of tangents at the end points of the chords of a circle which passes through a fixed inner or outer point of the circle, is called polar line and the fixed point is called pole with respect to the circle. If there are two points p and a such that polar line of p passes through a and polar line of a passes through p then the two points p and a are called conjugate points to each other.

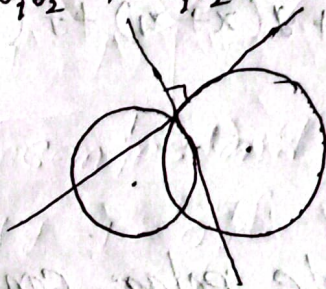


13. Orthogonal Circle: If two circles intersect at right angle i.e. the tangent at their point of intersection are at right angle then the circles are called orthogonal circles.

The circles $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$

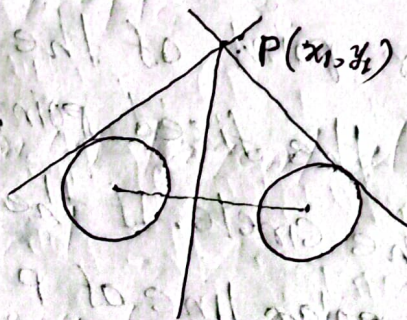
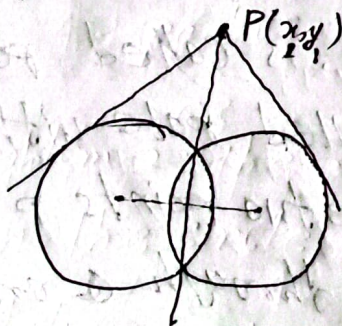
& $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$

are orthogonal if $2g_1g_2 + 2f_1f_2 = c_1 + c_2$



14. Radical axis:

The locus of a point which moves so that the lengths of tangents drawn from it to the two circles are equal, is called radical axis of the two circles.



$$S_1 = x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$$

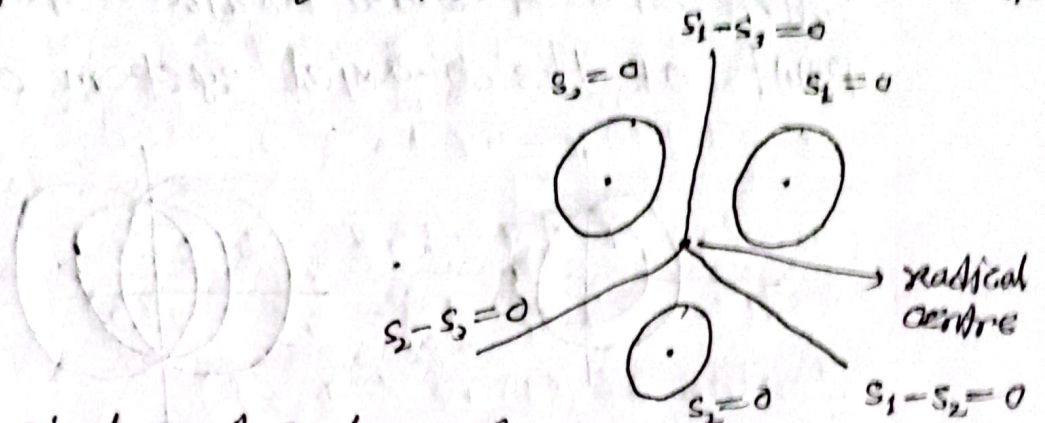
$$S_2 = x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$$

From the definition,

$$\Rightarrow \sqrt{x_1^2 + y_1^2 + 2g_1x_1 + 2f_1y_1 + c_1} = \sqrt{x_1^2 + y_1^2 + 2g_2x_1 + 2f_2y_1 + c_2}$$

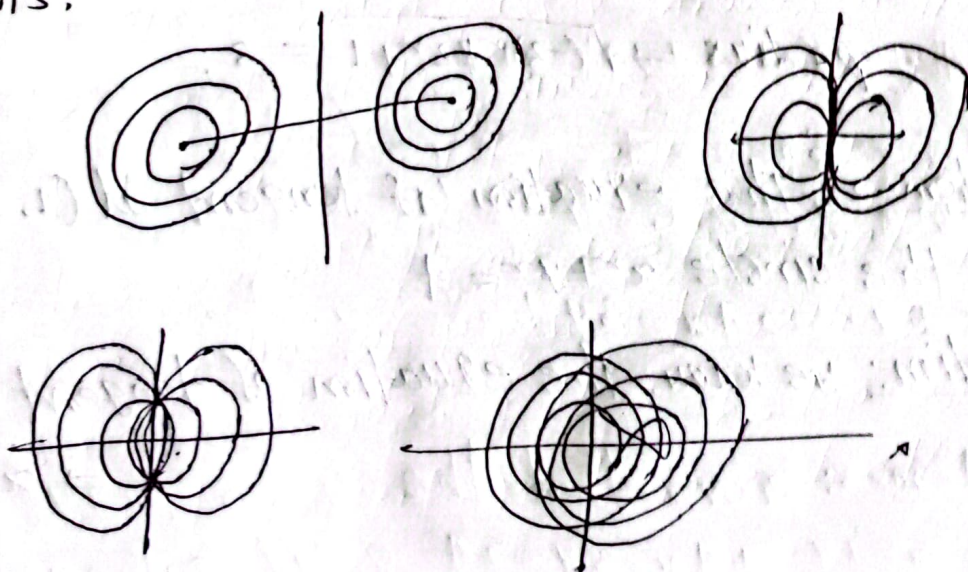
$$\therefore 2(g_1 - g_2)x_1 + 2(f_1 - f_2)y_1 + c_1 - c_2 = 0 \Rightarrow S_1 - S_2 = 0$$

15. Radical centre: The point of concurrence of the radical axes of each of the three circles whose centres are not collinear is called the radical centre.

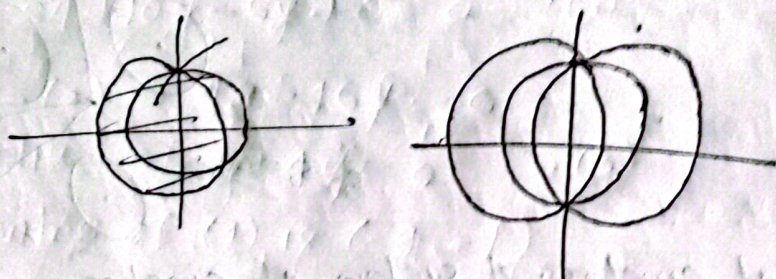


16. Co-axial circles: A system of circles in which every pair of circles has the same radical axis is called co-axial system.

The centres of circles of a co-axial system must lie on a line perpendicular to the common radical axis.



17. Limiting points of a co-axial system of circles;
The centres of the circles of a co-axial system which are of zero radius are called the limiting points of the co-axial system of circles.



Problem-1: Find the centre and radius of the circle $x^2 + y^2 + 4x - 4y - 1 = 0$.

Solution: The given circle is

$$x^2 + y^2 + 4x - 4y - 1 = 0 \longrightarrow (1)$$

Here, Centre $(-2, 2)$

$$\text{radius} = \sqrt{(-2)^2 + 2^2 + 1} = 3.$$

Problem-2: The equation of tangent at $(2, 2)$ lying on the circle $x^2 + y^2 = 4$

Solution: We know, the equation of tangent $xx_1 + yy_1 = r^2$
 $\Rightarrow x + 2y = 4.$

Problem-3: Show that the straight lines $y = mx \pm a\sqrt{1+m^2}$ are always tangents to the circle $x^2 + y^2 = a^2$.

Solution: Let. $y = mx + c \rightarrow (1)$

meet the circle $x^2 + y^2 = a^2 \rightarrow (2)$

From (1) & (2), we get

$$x^2 + (mx + c)^2 = a^2$$

$$\Rightarrow x^2 + m^2x^2 + 2mcx + c^2 = a^2$$

$$\Rightarrow (1 + m^2)x^2 + 2mcx + c^2 - a^2 = 0$$

If the line (1) always touches the circle (2) then x has equal roots in (2)

$$\therefore (2mc)^2 - 4(1+m^2)(c^2 - a^2) = 0$$

$$\Rightarrow m^2c^2 - c^2 + a^2 - m^2c^2 + m^2a^2 = 0$$

$$\Rightarrow c^2 = \pm \sqrt{1+m^2} \cdot a \rightarrow (3)$$

using (3), from (1)

$$\therefore y = mx \pm a\sqrt{1+m^2} \quad (\text{shown}).$$

Problem-4: Find the equation of a circle which passes through the points $(3, 5)$ and $(5, -3)$ and has its centre on the line $2x + y = 27$.

Solution: Let us consider the given circle

$$x^2 + y^2 + 2gx + 2fy + c = 0 \rightarrow (1)$$

(1) passes through the points $(3, 5)$ and $(5, -3)$. we get

$$\therefore 3x^2 + 5y^2 + 6x + 10y + C = 0$$

$$\Rightarrow 34 + 6x + 10y + C = 0 \longrightarrow (2)$$

$$\text{and, } 5x^2 + (-3)y^2 + 10x - 6y + C = 0$$

$$\Rightarrow 34 + 10x - 6y + C = 0 \longrightarrow (3)$$

$$(3) - (2) \Rightarrow 4x - 16y = 0$$

$$\Rightarrow x - 4y = 0 \longrightarrow (4)$$

Centre of (1) is $(-g, -f)$

$\therefore$ centre on the line $2(-g) + (-f) = 27$

$$\Rightarrow 2g + f = -27 \longrightarrow (5)$$

$$\text{using (4), from (5)} \Rightarrow 2 \cdot 4f + f = -27$$

$$\Rightarrow f = -3$$

$$\therefore (4) \Rightarrow g = -12$$

putting g & f from (2), we get

$$34 + 6(-12) + 10(-3) + C = 0$$

$$\Rightarrow C = 68$$

putting g, f & C , from (1) $\Rightarrow$

$$x^2 + y^2 - 24x - 6y + 68 = 0. \underline{\underline{Ans}}$$

Problem-5: Prove that the circles $x^2 + y^2 - 3x + 8y - 2 = 0$ and $x^2 + y^2 + 4x - 5y - 24 = 0$ cut orthogonally.

Solution: The given equation is

$$x^2 + y^2 - 3x + 8y - 2 = 0 \longrightarrow (1)$$

$$\therefore 3x^2 + 5y^2 + 6x + 10y + C = 0$$

$$\Rightarrow 34 + 6x + 10y + C = 0 \longrightarrow (2)$$

$$\text{and, } 5x^2 + (-3)y^2 + 10x - 6y + C = 0$$

$$\Rightarrow 34 + 10x - 6y + C = 0 \longrightarrow (3)$$

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Problem-5: Prove that the circles $x^2 + y^2 - 3x + 8y - 2 = 0$ and $x^2 + y^2 + 4x - 5y - 24 = 0$ cut orthogonally.

Solution: The given equation is

$$x^2 + y^2 - 3x + 8y - 2 = 0 \longrightarrow (1)$$

Centre $(3/2, -4)$

$$\text{Radius} = \sqrt{(3/2)^2 + (-4)^2 + 2} = \frac{9}{2}$$

$$\text{and. } x^2 + y^2 + 4x - 5y - 24 = 0 \longrightarrow (2)$$

Centre $(-2, 5/2)$

$$\text{Radius} = \sqrt{(-2)^2 + (5/2)^2 + 24} = \sqrt{\frac{137}{4}}$$

Distance of two centre, $C_1 C_2$

$$= \sqrt{\left(\frac{3}{2} + 2\right)^2 + \left(\frac{5}{2} + 4\right)^2} = \sqrt{\frac{218}{4}}$$

$$\therefore C_1 C_2^2 = \frac{218}{4}$$

$$\text{We know, } C_1 C_2^2 = r_1^2 + r_2^2 = \left(\frac{9}{2}\right)^2 + \left(\frac{137}{4}\right)^2 = \frac{218}{4}$$

(proved)

Problem-6: Find the co-ordinates of the limiting point of the co-axial system determined by circles $x^2 + y^2 + 4x + 2y + 5 = 0$ and $x^2 + y^2 + 2x + 4y + 7 = 0$.

Solution: Given that,

$$x^2 + y^2 + 4x + 2y + 5 = 0 \longrightarrow (1)$$

$$\text{and } x^2 + y^2 + 2x + 4y + 7 = 0 \longrightarrow (2)$$

The radical axis of two given circles is

$$(x^2 + y^2 + 4x + 2y + 5) - (x^2 + y^2 + 2x + 4y + 7) = 0$$

$$\Rightarrow 2x - 2y - 2 = 0$$

$$\Rightarrow x - y - 1 = 0$$

Hence the equation of any circle co-axial with the given circles is

$$x^2 + y^2 + 4x + 2y + 5 + \lambda(x - y - 1) = 0$$

$$\Rightarrow x^2 + y^2 + x(4 + \lambda) + y(2 - \lambda) + (5 - \lambda) = 0$$

$$\text{Centre } \left(-\frac{4+\lambda}{2}, -\frac{2-\lambda}{2}\right)$$

Radius of this circle

$$= \sqrt{\left(-\frac{4+\lambda}{2}\right)^2 + \left(-\frac{2-\lambda}{2}\right)^2 - (5-\lambda)}$$

Thus the radius is zero

$$\sqrt{\left(-\frac{4+\lambda}{2}\right)^2 + \left(-\frac{2-\lambda}{2}\right)^2 - (5-\lambda)} = 0$$

$$\Rightarrow (4+\lambda)^2 + (2-\lambda)^2 - (5-\lambda)4 = 0$$

$$\Rightarrow \lambda^2 + 8\lambda + 16 + 4 - 4\lambda + \lambda^2 - 20 + 4\lambda = 0$$

$$\Rightarrow 2\lambda^2 + 8\lambda = 0 \Rightarrow \lambda = 0, -4$$

putting $\lambda = 0, -4$; Centre $\left(-\frac{4+0}{2}, -\frac{2-0}{2}\right)$ and

$$\left(-\frac{4-4}{2}, -\frac{2+4}{2}\right) = (-2, -1), (0, -3). \underline{\underline{Ans}}$$

Problem-7: Find the equation of the circle whose diameter is the common chord of the circle

$$x^2 + y^2 + 2x + 3y + 1 = 0 \text{ and } x^2 + y^2 + 4x + 3y + 2 = 0$$

Solution: The circle whose diameter is the common chord of the circle

$$S_1 = x^2 + y^2 + 2x + 3y + 1 = 0 \longrightarrow (1)$$

$$S_2 = x^2 + y^2 + 4x + 3y + 2 = 0 \longrightarrow (2)$$

The radical axis of (1) and (2) is

$$(x^2 + y^2 + 2x + 3y + 1) - (x^2 + y^2 + 4x + 3y + 2) = 0$$

$$\Rightarrow 2x + 1 = 0$$

Any circle passing through the point of intersection of the two circles is $S_1 + \lambda S_2 = 0$

$$(x^2 + y^2 + 2x + 3y + 1) + \lambda (x^2 + y^2 + 4x + 3y + 2) = 0 \rightarrow (3)$$

$$\Rightarrow (1 + \lambda)x^2 + (1 + \lambda)y^2 + 2(1 + 2\lambda)x + 3(1 + \lambda)y + (1 + 2\lambda) = 0$$

$$\Rightarrow x^2 + y^2 + \frac{2(1 + 2\lambda)}{1 + \lambda}x + \frac{3(1 + \lambda)}{1 + \lambda}y + \frac{1 + 2\lambda}{1 + \lambda} = 0$$

$$\Rightarrow x^2 + y^2 + \frac{2(1 + 2\lambda)}{1 + \lambda}x + 3y + \frac{1 + 2\lambda}{1 + \lambda} = 0$$

$$\therefore \text{Centre} \left(-\frac{1 + 2\lambda}{1 + \lambda}, -\frac{3}{2} \right)$$

As the centre lies on $2x + 1 = 0$

$$\therefore 2 \left(-\frac{1 + 2\lambda}{1 + \lambda} \right) + 1 = 0 \Rightarrow \lambda = -\frac{1}{3}$$

Putting $\lambda = -\frac{1}{3}$, from (2) $\Rightarrow$

$$(x^2 + y^2 + 2x + 3y + 1) + \left(-\frac{1}{3}\right)(x^2 + y^2 + 4x + 3y + 2) = 0$$

$$\Rightarrow 3x^2 + 3y^2 + 6x - 9y + 3 - x^2 - y^2 - 4x - 3y - 2 = 0$$

$$\Rightarrow 2(x^2 + y^2) + 2x + 6y + 1 = 0 \quad \underline{\underline{\text{Ans}}}$$